

MAKING & FINING CYDER

Timothy Matlack of Lancaster, 1807-1808

Two papers, transcribed in full, with the method extracted · compiled 13 August 2026

What this is

Timothy Matlack published on cider twice. Both papers sit in the same book — volume I of the *Memoirs of the Philadelphia Society for Promoting Agriculture*, printed in Philadelphia by Jane Aitken in 1808. Neither was written as a manual. Both are private letters that the Society printed.

The two papers

- **pp. 109-118** — “Account of a new Pummice Press, with some remarks upon Cyder making.” Two letters to Richard Peters, President of the Society, dated Lancaster 7 and 27 February 1807; read to the Society 10 March 1807. Engraved plate of the press facing p. 115, keyed description pp. 116-118. Despite the title, this is the paper with more cider thinking in it.
- **pp. 268-273** — “Observations on making and fining Cyder, and on Peach Trees.” A letter to Dr. James Mease, dated Lancaster 7 March 1808; read the following day. Runs on to p. 279 on peach trees, not transcribed here.

Why he nearly didn't write it

Both papers open with the same reluctance. Cider making depends on fermentation, which he calls a subject less understood than any other to which philosophy or chymistry have attended. Pennsylvania's climate is too variable for directions to hold. And he expected the blame when other men followed his practice and failed. What talked him into it was age: to reason against fixed prejudices, he writes, is folly that ought not to be expected beyond the age of 70 — and then he does it anyway. By 1807 he had been trying apples for cider more than forty-eight years.

How to use this document

- Part One is the method, pulled out of both papers and put in working order. Read this if you want to know what he actually did.
- Part Two is the source note — what is contested, what Coelho's biography gets slightly wrong, and where to read the originals.
- Appendices A and B are the two papers transcribed in full, original spelling kept.

PART ONE

The method

Assembled from both papers and ordered as the work happens. Every step is Matlack's; nothing has been supplied from other period sources.

Mill and press (from the 1807 paper)

- Press immediately from the mill. The best ciders he ever saw, if not every truly excellent one, were pressed straight from the mill — paler cider, more perfectly bright in the cask, less lees in the bottle. The Society appended a footnote noting this runs against general practice and urging farmers to test it.
- If the pommace does stand, never measure the wait in hours. Judge by the heat of the air: the same interval that does nothing in one season will start an acid fermentation in a warmer one.
- Lattice the floor, then hair cloth or coarse bagging over it and on three sides of the crib, so the must runs off quite fine.

Temperature and the cap (from the 1807 paper)

- His temperature evidence comes from beer, not cider. Malt wort fermented at 65° and drawn off its yeast in time came spontaneously fine and even perfectly bright, soft and free of bitterness. The same wort at 76° fixed a cloud no art could remove, decomposed enough for the hop resin to offend the palate, and grew steadily more bitter until it reached the bitter of aloes.
- Warmth is the first sensible effect of fermentation: it expands the air in the vesicles of the pulp and floats them up. Remove them when they rise — their return increases the fermentation.
- The pulp of the apple forms part of cider's yeast. Left in, it decomposes the cider and puts the precise bitter of an apple leaf on the palate, before the acetous fermentation even begins.
- “Bright” is a brewer's term, and he uses it deliberately: not merely fine, but the perfect transparency in which liquors are tasted in their purity.

On the gauge

- He gives the same 11-to-24 spread in both papers — Vandever against Cooper's sweet russet — and in 1808 supplies the unit: pennyweights heavier than a pint of rain water. Cooper's sweet russet is the richest must of every apple he examined across more than forty-eight years; the house apple is next heaviest.

- He wanted farmers using Dicas's hydrometer for this and said they would hardly credit the difference until they did. A Society footnote supplies the instrument's name: saccharometer.

Ferment (from the 1808 paper)

- Take the Virginia crab if you can get it. Light must, superior cider, and — the reason Matlack rated it above every other apple — far less tendency to run away in fermentation, so it needs less judgement to restrain.
- Weigh the must: a pint of juice against a pint of rain water, difference in pennyweights. Vandever came in at 11 dwt heavier, Cooper's sweet russet at 24 dwt, in the same season. Dry seasons give a heavier must, wet seasons lighter.
- Ferment by the weather, watching every change and especially any swing from cool to warm. He would not give precise directions: the climate is too variable and the best cellars run too warm in cider season.
- To restrain too rapid a fermentation: rectified spirits of wine, burnt taste destroyed by powdered orris root at one ounce to the pint. One spoonful per hogshead is the safe maximum at any one time, laid on the surface with the cask full to the bung hole.

Fine (from the 1808 paper)

- There must be no fermentation whatever at the time of fining. That means before the spring fermentation begins, generally about the last of March — or after it ceases, by which time the cider has usually gone acid enough not to be worth fining.
- Common staple isinglass, about five staples to two hogsheads. Pound it and break it into threads.
- Dissolve it in the cider to be fined: beat that cider well, several times a day, for two or three days, then strain through a flannel bag.
- Best general practice — pour the fining into the empty cask, then draw the cider off onto it. This leaves much of the sediment behind, checks insensible fermentation, and mixes cider and fining intimately.
- Fill quite full, pour the spirits of wine on the surface. Fine and bright in six to eight days; draw off immediately and bung close or bottle. Under-fermented cider will break your bottles.

Seal

- Bung close and wax over the bung. Drive the bung home, then bore a gimlet hole beside the bung hole and leave it open while you cement over the bung.

- Without the vent, the warmth of the cement expands the air below, blisters up through the air hole, and defeats the seal. Once the cement has cooled and hardened, drive a square white oak plug into the gimlet hole.

Three warnings

- Too much is expected of cider. The best Madeira will not keep on less than eight gallons of brandy in the hundred, twelve more commonly — so unaided apple juice cannot be expected to stand through a Pennsylvania summer.
- Don't fortify. Brandy blends invisibly into wine over years; cider exposes even the smallest quantity of the best spirits more readily than any other liquor.
- Don't boil the must down. Concentrating by boiling makes added yeast necessary, and he never met a boiled one his stomach didn't complain of. Concentrate by frost instead, on cider already duly fermented — that gives a delicate bottled liquor that stands for years and improves with time.

PART TWO

Source note

Where to read the originals

- Internet Archive, free and complete:
archive.org/details/memoirsofphilade1808phil — scanned from the New York Botanical Garden copy, 567 pages, full OCR. The reader takes printed page numbers directly.
- The press paper begins at printed p. 109; the fining paper at p. 268.
- HathiTrust catalog record 100112068 covers volumes 1–5 (1808–1826).
- The Society deposited its records with the University of Pennsylvania in 1888 and still exists. Physical papers: Kislak Center, PSPA Records, Ms. Coll. 92.

The wrinkle in Coelho's citation

Chris Coelho's biography of Matlack is what sends most readers looking for this. Chapter Three, note 35 reads: Timothy Matlack, "Observations on making and fining Cyder, and on Peach Trees," *Memoirs of the Philadelphia Society for Promoting Agriculture* v. 1 (1808): 112.

The title belongs to the 1808 paper, which begins at page 268. The page number falls inside the 1807 press paper, which runs 109–118. Coelho appears to have merged the two. Cite them separately.

Chapter Three, note 34 gives the source for the advertisement quoted on his page 33: Timothy Matlack, advertisement, *Pennsylvania Chronicle* 3, no. 36, October 2, 1769. Worth flagging, because the narrative places that advertisement immediately after a paragraph about 1764 and reads as though it belonged to that year.

The stipulation in the advertisement is the interesting part — he wanted cider delivered before it began to ferment, because he intended to run the fermentation himself. That is the same expertise, thirty-eight years earlier.

What the Society disputed

Matlack's claim that the best ciders are pressed immediately from the mill did not go unchallenged. A member appended a footnote to the same page noting the opinion is so different from what cider makers generally hold that experiments are well worth making, and recommending the subject to farmers. That disagreement is printed on the page, and it is still the live question in the paper.

A modern gloss on the pennyweights

This conversion is not Matlack's. A pennyweight is one twentieth of a troy ounce, about 1.56 grams. Against a wine pint of water at roughly 473 grams, 11 dwt heavier lands near 1.036 specific gravity and 24 dwt near 1.079 — in modern terms, roughly 4.5 and 10 per cent potential alcohol. Treat as approximate: the exact pint measure he used is not stated.

Where this sits in the literature

Matlack's papers predate what is usually called the first American cider manual, and are among the earliest American cider-making instructions in print.

- **1793** — A. Crocker, "A Practical Essay on Raising Apple Trees, and Making Cider," *Memoirs of the American Academy of Arts and Sciences*, vol. 2, no. 1, pp. 100-113
- **1807-08** — Matlack, the two papers transcribed here
- **1817** — William Coxe, *A View of the Cultivation of Fruit Trees, and the Management of Orchards and Cider* (Philadelphia: M. Carey). Usually credited as the first American cider manual. Coxe was a member of the same society and appears in the 1808 volume.

English precursors he would have known

- John Evelyn, *Pomona*, appended to *Sylva*, 1664
- John Worlidge, *Vinetum Britannicum*, 1676
- Hugh Stafford (attributed), *A Treatise on Cyder-Making*, 1753
- Thomas Chapman, *The Cyder-Maker's Instructor*, 1757
- Thomas Andrew Knight, *A Treatise on the Culture of the Apple and Pear, and on the Manufacture of Cider and Perry*, 1797; 2nd ed. 1801

Ruled out along the way

Recorded so the same ground doesn't get covered twice.

- **Memoirs volume 3 (1814)**. Booksellers list this volume as "by Timothy Matlack" because his paper on cultivating the vine is item I in its contents. He is not the author of the volume, and its cider content is by others.
- **Henry Wynkoop**. Wrote on cider from the Hewes (Hughes) crab in that 1814 volume. Not Matlack.
- **Timothy Pickering**. Contributed an account of the Virginia crab to the same volume — same first name, adjacent subject, easy to confuse in a table of contents.

- **The 1780 oration.** Matlack's address to the American Philosophical Society, 16 March 1780, printed by Styner and Cist, is about agriculture generally. No cider.

APPENDIX A

The press paper, in full

Pages 109–118. Transcribed from the 1808 scan; original spelling and punctuation kept. Footnotes are marked in the text where they fall on the page.

Pages 109–118, read to the Society 10 March 1807. Two letters to Richard Peters, the Society's President, plus a keyed description of the engraved plate. More of Matlack's cider thinking is here than in the better-known 1808 paper.

***Account of a new Pumice Press, with some remarks upon Cyder making.
By Timothy Matlack, of Lancaster.***

Read March 10th, 1807.

Lancaster, February 7th, 1807.

Sir,

Colonel Johnston of your city paid me a visit, and I shewed him, as I had done some others, a pumice press that I had made for my own use; intended principally for making of wines from currants, black berries, grapes and other small fruits; but as I wished to make wine from quinces which is beyond question, little if any inferior to that of the best grape, and also, expecting to make some perry, it seemed best to extend its size, to those objects, especially as the encreased expence would be very small. I therefore fitted it to those objects, and as it now appears, to that of cyder making, in a way far indeed beyond my first intention. On viewing it, Colonel Johnston suggested the idea of sending a sketch of it to you, assuring me that it would not fail of a favourable reception; and I now enclose a side view of it, that will shew the principle on which it is constructed; and I trust, demonstrate that it is capable of an almost incredible force, within a small space, by very simple means, and at a very small expence; and also that it can be used with the greatest facility, and when done with for the season, can be laid securely by, without occupying much house room.

Several persons who have seen the press have expressed their idea, from the smallness of the cribb, that it was intended only for a model; not adverting to the space left for a much larger cribb; nor instantly perceiving that both levers, acting wholly within the machine, press with equal force, both upwards and downwards; but no one who has examined it, has failed to express his opinion of its promising fair to become really useful; and if it shall prove to be so, it will afford me ample satisfaction for my trouble.

Be this as it may, I am convinced that the best chance for becoming so, will be derived from your care and attention, to which it is committed with the greater pleasure, as I confide that you will allow me the credit of a respectful attention.

I am your most obedient, humble servant,

T. MATLACK.

Hon. Richard Peters Esq., President, Agric. Society, Philad.

[Mr. Matlack having been requested, by order of the society, to procure a model to be made, and to transmit it with further explanations, was so obliging as to comply with that request; and the model was accompanied by the following letter.]

Lancaster, February 27th, 1807.

Dear Sir,

The model now sent you under care of Col. Johnston, is on a scale of an inch to a foot.—The levers to press 40 for 1, and the cribb to contain 40 bushels, which I think may be wrought at least three times, while one of 80 bushels can be wrought once in the common mode.

I have no wish to engage in the question of cyder making, further than to suggest this mode of simplefying the lever; the sole inconveniency of which appears to be the frequency of removing the weight; which from the unalterable law of the lever, must be proportioned to the increase of pressure. Hence each weight should be no greater than is within the strength of the attendant; or, which is the same thing, the weight of each lever should be divided for that purpose. It is planned for three pair of levers, of which two only are inserted, and the space for the third blocked; either of which a stout lad of twelve years old may handle. It is intended, that two of the three should continue to press while the other is raised; in doing of which an inch board of a foot width, and of a proper length, will be quite sufficient to support the first lever: or with a little more strength it may be turned over, out of the way. As to the second lever, it can be withdrawn, and replaced in less than a minute. But enough of the model, which it was more trouble to make than the working press, rough as it is. I chose to make it myself, rather than employ a mechanic here, because I well know, that it would require more time to get any thing done by them, than to make it, if I was able.

To reason against fixed prejudices is folly that ought not to be expected beyond the age of 70: it always gives offence, and is generally fruitless. Yet, lest it may look like sneaking from the question you suggest (with your usual address) I will venture to say the best cyders that I have ever seen, if not all the truly excellent, has been pressed immediately from the mill.

[Society's footnote: This opinion is so different from that generally entertained by cyder makers, that experiments are well worth making to determine the point, or to ascertain the difference which pressing the pummice immediately from the mill, and permitting it to remain some hours before pressing, would occasion in the quality of the liquor. The subject is earnestly recommended to the attention of farmers.—Note by a Member.]

The truth is, that cyder making depends on fermentation; a subject less understood than any other to which philosophy or chymistry, have attended; and my

knowledge of it, is just sufficient to warrant the sentiment, and to have learned, that the little that is known on the subject, it is extremely difficult to communicate, or to reduce to practice, in a country whose climate is so extremely variable as that of Pennsylvania, sometimes even in the cyder making season, so warm as to put the fermentation above controul; and at times soon after, so cold as totally to suspend it; so that it unavoidably commences again and goes beyond its proper point in the spring.

A wort of malt and hops, fermented at 65° and separated from its yeast in due time, becomes spontaneously fine, and even perfectly bright; is a fine colour according to the colour of its materials; is soft and free from bitterness. A part of the same wort fermented at 76° has a cloud fixed in it, which art has not yet been able to remove; is so far decomposed as to cause the resin of the hop to offend the palate with its bitter, which grows more and more offensive by time and finally acquires the offensive bitter of the aloes.

The pulp of the apple forms at least a part of the yeast of cyder, and if not separated by fermentation, but suffered to remain, will decompose the cyder, and exhibit to the palate the precise bitter of the apple leaf, previously to the commencement of the acetous fermentation. Warmth is the first sensible effect of fermentation. This expands the air contained in the vesicles of the pulp, and occasions them to rise; they should then be removed; their return increases the fermentation.

Our farmers have not yet attended to the important fact of difference in the strength and weight of the must from the different kinds of apple, on which the successful practice of fermentation will forever depend—And which they will hardly credit until the use of Dica's hydrometer, or some such instrument, finds its way amongst them. To you, I may venture to say, that by even a more accurate mode of determining this difference, beyond the weight of rain water, I have found it to be so incredibly great as 11 to 24, which, I think (for my notes on this subject are in the city) was between the juice of the Vandever and of Cooper's sweet russett, which produces the richest must of all the apples I have examined, and I have tried very many for more than 48 years back; the next heaviest is the house apple.

[Society's footnote: The appropriate name of this valuable hydrostatic instrument I do not recollect. (It is called "Saccharometer.")]

Having said thus much, it would be wrong not to add, that the Virginia crabapple affords a juice extremely different from that of any other apple I know of, and appears to be less liable to an excess of fermentation, the base of our common cyder, than that of any other apple. The cause of this difference, I am quite willing to leave others to guess at, or to enquire concerning by more rational means, at their choice.

For the truth of this important fact you may venture to take my assurance: to wit—That the sooner the pummice is pressed after grinding, the paler the cyder will be—the more perfectly bright it may be made in the cask—and the less lees it will deposit in the bottle. A moment's reflection will satisfy you, of the incorrectness of the practice of measuring the length of time which pummice should remain after grinding, and before it is put to press, by hours, without regard to the heat of the air at the time. You will perceive, that one season the same length of time will produce no sensible effect, which at a much warmer season would induce the commencement of an acid fermentation.

[Matlack's footnote: The word bright is a term used by brewers to express the difference between what is commonly called fine, and that perfect transparency in which liquors are, alone, tasted in their purity.]

Having gone so much further on this subject, than I had intended, I cannot help asking myself the question, ought I to ask your pardon for it, or my own? Perhaps the answer should be, that I deserve it from neither. However I am certain of this—that I am with much esteem and respect,

Your most obedient and very humble servant,

T. MATLACK.

Hon. Richard Peters, President Agric. Soc. Philad.

The press, from the plate description (pp. 116-118)

- Crib 21 by 20 inches, 20 inches high — four bushels. But the press is equal to a crib of 48 by 48 inches and 36 inches high, holding 37 bushels.
- A 100 lb weight, geared as five times five to one, presses 2,500 lb downward. Because the uprights are fastened to the side planks, each lever's toe bears the crib upward with the same power that the heel presses down — so the actual pressure on the pommice is 5,000 lb.
- Two compound levers side by side give 10,000 lb. Lengthen the supporting pin by five inches and two more levers can be added outside the uprights for another 10,000 lb, “and so infinitely.”
- Iron pin at A is 1½ inches thick, passing through both uprights. Holes B, B, B let the lever be moved to any required height. Side plank 7 ft 4 in from out to out; lever arms 6 ft 8 in, five times the length below the fulcrum.
- Floor perforated with two augur holes of 1¼ inch. Lattice bottom on three ribs 1¼ in wide and ½ in thick, hair cloth or coarse bagging over it and on three sides of the crib, so the must runs off quite fine.
- Three sides of the crib tenon into the plank side with headed pins, knocked out when the pommice is pressed; that side turns outward, the pommice is thrown out, and the crib is returned and refilled.

- Two days' work for a carpenter, tested not guessed: an apprentice helped saw plank for one day and Matlack finished it on the second. Plank 2½ in thick cost \$2.50 delivered; a chain, a pin and a hundred 4d cut nails are the whole of the iron work. A strong withe will do for the chain and tough hickory for the pin.
- Whole press occupies 13 ft 7 in. It comes apart in about 20 minutes, goes back together in under an hour, and packs into a box 20 inches square and 8 feet long.

APPENDIX B

The fining paper, in full

Pages 268–273. The paper continues to p. 279 on peach trees; that section is summarised at the end rather than transcribed.

Read here off the 1808 scan, pages 268–273. Original spelling and punctuation kept. The paper runs on to page 279 with a long section on peach trees, summarised at the end.

Observations on making and fining Cyder, and on Peach Trees. By Timothy Matlack, Esq.

Read March 8th, 1808.

Lancaster, 7th March, 1808.

Dear Sir,

A visit a few days ago from the reverend doctor Muhlenbergh of this borough, and a communication of a paragraph of your letter to him, for which I thank you, brings in review your letter of last fall, which came to hand when there was little probability of my ever being able to acknowledge the receiving of it.

The error in the size of Mr. Cooper's vine has happened on the most favourable side, and it is fortunate that it is so: for the measure, as given, is quite large enough to obtain credit. He has little reason to thank me for the publication, since it has occasioned him so many troublesome visits, from persons who sought only to gratify an idle curiosity, and he has no other compensation to hope for, than the pleasure it must afford him to see, that it has also drawn the attention of some gentlemen whose object is public improvement, and that it has tended to encourage them in their laudable pursuits.

The making and fining of cyder, so as to produce that excellent liquor in the perfection it is capable of, would no doubt be a great public benefit, and I confess that I once thought seriously, of publishing the observations I had made on that subject; but on considering the fixed prejudices which a performance of that kind would have to contend against; that the success depends mainly on fermentation, the theory of which, you know, is less understood than any other branch of chymical science, and consequently the great difficulty of communicating intelligibly, what little I knew on the subject in practice, I was deterred from an attempt which promised so little advantage to any body, and threatened so much vexation to myself, from the blame which want of success in those who might pretend to have followed the practice I should recommend, would perhaps, but unjustly, being upon me. It looks very like vanity to say, that I knew too much of the matter to hope for success in the undertaking.

But, in support of this opinion and to justify it, permit me to say, that I knew there was a much greater difference in the must of cyder than would be credited by our cyder makers; and that the degree of fermentation each would bear depended on its degree of strength, and that, therefore, there was very little probability that they would succeed under any possible directions I could give: for example, I knew that a pint of Vandever juice weighed but eleven penny weights in a pint, more than rain water, when in the same season a pint of juice from Cooper's sweet russett, weighed twenty-four penny weight heavier than the same water; and I knew also that the juice of the same kind of apples differed greatly in its specific gravity in different seasons, dry seasons producing a heavier and wet seasons a lighter must.

These facts led me to suppose, that its strength depended on the quantity of saccharine matter contained in it, and that consequently its specific gravity would shew its real strength, and my practice was founded on this supposition and very generally was attended with considerable success; but, meeting with the must of the Virginia crab, so famous for its cyder, I learned that this was, at least, an exception to that principle; for its specific gravity was, as near as I remember (for my notes are in the city) rather below the mean weight of our common cyder apples; and its cyder is not below our best cyders. This taught me that there was some other principle, less open to detection, on which the excellency of cyder must depend. I however proceeded in my usual mode of fermentation, and soon found that this must had much less a tendency to extreme fermentation, than that of our common apples of equal weight; and consequently, required less judgment to restrain it than others. The importance of this fact needs no comment: It gives a decided superiority to that apple for cyder, above all others.

The cyder you mention as so much approved by the French minister, was from the Virginia crab. I think in the year 1777 or '78. Cyder being then very scarce in Philadelphia, I had obtained but a single hogshead, directly from the press, and the fermentation was conducted with more than common care, and consequently became spontaneously not merely fine, but perfectly bright, and exhibited that appearance of bounding up of small drops to the height of six or eight inches, so highly pleasing in the finest Champaign wine, without any appearance of froth on the top of the glass, and retained all the delicacy of flavour which distinguishes that apple, free from the slightest degree of acidity.

In conducting this fermentation attention was paid to every change in the weather; especially where the change was from cool to warmer degrees. You will see then, how difficult it must be to give precise directions for conducting so nice a process, in a climate so extremely variable as ours, and where our best cellars are too warm, in the cyder making season, to conduct it with certainty of success in the best of them; and may conceive how reluctantly it will be undertaken by any one who has feeling enough to be careful to avoid even unmerited censure. The too rapid fermentation I have restrained, by a small quantity of rectified spirits of

wine, the empyreuma or burnt taste of which had been destroyed by powdered orris root, (an ounce to a pint.) A single spoonful to a hogshead is as much as can be used with safety, at any one time, and that should be laid on the surface of the cyder when the cask is full to the bung hole.

Where no great error in the fermentation has been committed, the fining of cyder is a very plain business; but it is indispensably necessary, that at the time of fining, there should not be the least degree of fermentation; of course, it must be done before the spring fermentation commences, (which generally happens about the last of March,) or be delayed until that fermentation ceases, at which time, it most frequently has acquired a degree of acidity, that renders it not worth fining. The common staple isinglass, is, perhaps, the safest fining; about five staples to two hogsheads. It is to be dissolved in the liquor intended to be fined, after being pounded and broken into threads. To dissolve it completely, it is necessary to beat the cyder containing it well, several times a day, for two or three days, and then to strain it through a flannel bag.

The best general practice is, to pour your fining into the empty cask, and then draw off your cyder and pour it on the fining. This leaves behind, a great part of the sediment, checks insensible fermentation, and mixes intimately the cyder with the fining. Then the cask being quite full, pour on the spirits of wine, on the surface. It will generally become quite fine and bright in six or eight days, and should then immediately be drawn off, and bunged up close, or bottled. But if it has not been sufficiently fermented, it will break your bottles.

If drawn into casks they should be bunged close, and waxed over the bung to keep the air entirely out. To do this effectually, after the bung is carefully driven in, you must bore a gimblet hole near the bung hole, and leave it open until you have covered your bung with the cement; otherwise you will cover the bung, and leave open the small holes on the side of the bung; the warmth of the cement increasing the quantity of the air below, will throw up a blister through the air hole, and forever disappoint the attempt to close it. The gimblet hole admitting the warm air to pass, the cement keeps its place and closes the aperture, and when the cement is cooled and hardened, the gimblet hole is completely stopped by driving a white oak square plug into it.

Another strong reason for declining the task, was, that too much is expected from cyder. The best Madeira wine will not keep with less than eight gallons of brandy in the hundred, and twelve is more commonly used: how then, can it be expected that the juice of the apple, without aid, should stand through our summer's heat? Brandy mixes its taste with the wine, in a course of years, so as not to be perceived on the palate; but cyder exposes even the smallest mixture of the best brandy, or any known spirits, more readily than any other liquor. The concentration of cyder by frost, in our coldest weather, if it has been previously duly fermented, affords a delicate bottled liquor, that will stand for years, and improve by time; but the concentration of the must, by long boiling, renders yeast absolutely necessary to

ferment it to a degree, suitable for drink at our tables; and I have never met with any, that my stomach did not complain of, after even a moderate draught of it: whether owing to its being boiled in copper vessels, or some other cause, I cannot venture to say.

[The paper continues "On Peach Trees" from page 273 to 279, and closes:]

Sensible of the great advantage to the public, to be derived from the exertions of the agricultural society, I wish the zeal of its members, may long continue to increase.

and am yours and their, most obedient servant,

TIMOTHY MATLACK.

Dr. James Mease.

Both cider papers transcribed from the 1808 Internet Archive scan, pages 109-118 and 268-273. Coelho endnotes transcribed from the printed book. The peach-tree section, pp. 273-279, is not transcribed. Pennyweight-to-gravity conversions are modern and approximate.